Bridging the Gap: A Heterogeneous Collaborative Learning Framework via Frequency-Aware Synergy for Multimodal Remote Sensing Semantic Segmentation

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***Abstract*—Multimodal remote sensing image segmentation remains challenging due to cross-modal information gaps, background noise interference, and the inherent limitations of single-architecture feature learning. To address these issues, we propose a heterogeneous collaborative learning (HetCoLearn) framework that facilitates robust feature interaction via frequency-aware synergy. HetCoLearn comprises two complementary networks: the frequency-domain Gaussian multimodal network (FGMNet), which integrates a Fourier fusion module to capture cross-modal frequency correlations and employs an attention-based Gaussian filter for noise suppression, and the multiscale wavelet-based Manhattan network (MWMNet), which utilizes wavelet-based feature extraction to capture hierarchical spatial details and a Manhattan spatial-correlation module for precise edge detection. A hierarchical collaborative strategy enhances knowledge transfer through three components: 1) frequency-spatial feature alignment, which synchronizes global frequency patterns with local wavelet components; 2) collaborative weight correction within attention optimization; and 3) a consistency-guided decoding constraint that enforces semantic agreement between heterogeneous branches. Extensive experiments on the Vaihingen and Potsdam datasets validate the effectiveness of the HetCoLearn framework. Our approach outperforms several state-of-the-art methods, demonstrating its efficacy and robustness in complex remote sensing scenarios. The code and results can be found at https://github.com/QshHub/HetCoLearn**

***Index Terms*—Collaborative learning, convolutional neural network (CNN), multimodal remote sensing image, semantic segmentation.**

I. Introduction

SEMANTIC segmentation is a core task in computer vision, serving as a critical bridge between low-level image processing and high-level scene understanding. Semantic segmentation of high-resolution remote sensing imagery plays a vital role in numerous practical applications. These include land-cover classification [1], [2], which is crucial for ecological assessment and resource management; urban planning [3], [4], which is used to map land-use patterns and monitor urban expansion; land planning [5], which facilitates the rational allocation of spatial resources; and road extraction [6], [7], which supports traffic network optimization and infrastructure development.

The rapid progress in deep learning has significantly advanced image semantic segmentation, and is largely driven by convolutional neural networks (CNNs). The introduction of the fully convolutional network (FCN) [8] enabled end-to-end dense pixel-wise predictions through convolutional operations. Subsequent innovations include the U-Net [9], which has a U-shaped architecture with skip connections to integrate multiscale features, thereby improving both semantic and detail perception, and an encoder–decoder model that enhances boundary accuracy [10]. Studies have proposed the use attention mechanisms [11], [12], dilated convolutions [13], [14], and dynamic convolutions [15] to extract richer features. However, conventional methods still face challenges in capturing long-range dependencies while preserving finer local details. Moreover, self-attention mechanisms suffer from high computational costs when processing high-resolution images owing to their quadratic complexity, and exhibit inadequate sensitivity to local structural variations.

Recent advances in semantic segmentation have produced highly accurate models, such as pyramid scene parsing networks (PSPNets) [16] and bilateral segmentation networks

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(BiSeNets) [17]. However, their high computational cost limits their practicality in many real-world applications. The need for efficient models on mobile platforms has driven the adoption of knowledge distillation (KD) [18], [19], [20], [21], [22], which is a technique that transfers knowledge from a large teacher model to a compact student model, allowing the student to match or even exceed the performance of the teacher. KD, which was proposed by Hinton et al. [23], initially used soft labels; subsequently, it evolved to enable more comprehensive knowledge transfer, leading to student models with better performances and generalizations than other lightweight methods [24], [25]. However, applying KD to remote sensing image segmentation presents specific challenges. Offline distillation relies on a fixed teacher, resulting in unidirectional knowledge flow and limited generalization [26], prediction-based methods are computationally inefficient [27], and feature-based approaches often suffer from training instability. Thus, developing specialized KD strategies for remote sensing imagery is essential for improving the efficiency and generalization of compact segmentation models.

This study proposes a heterogeneous collaborative learning (HetCoLearn) framework to balance the inherent tradeoff between multimodal fusion and cross-modal noise susceptibility. Multimodal networks can effectively leverage auxiliary modalities to supply fine details missing from optical images; however, they are inherently vulnerable to noise propagation caused by sensor anomalies or data irregularities in the auxiliary sources. Conversely, unimodal networks are immune to cross-modal noise, and their representational capacity is constrained by the complete absence of complementary viewpoint cues. HetCoLearn therefore introduces a heterogeneous pair comprising a frequency-domain Gaussian multimodal network (FGMNet) and a multiscale wavelet Manhattan network (MWMNet). Specifically, the FGMNet serves as a global multimodal integrator that fully exploits cross-modal spectral–elevation correlations, whereas the MWMNet acts as a dedicated unimodal expert that preserves clean spatial structures in the pure RGB input. The proposed three-stage collaborative learning framework establishes a virtuous cycle of collaborative interaction and mutual reinforcement among heterogeneous networks operating in different frequency domains. The MWMNet implicitly absorbs structural elevation knowledge via bidirectional distillation while avoiding contamination from raw nDSM noise, and the FGMNet refines its spatial feature representation by inheriting fine-grained local details from the MWMNet.

Our main contributions are summarized as follows.

1) We introduce FGMNet, which uses multimodal RGB and nDSM data as inputs, integrates cross-modal features through a Fourier fusion module, and suppresses noise using a Gaussian attention mechanism.
2) Owing to the limitations in exploiting multiscale features and target edge information from unimodal RGB data, we propose MWMNet with a wavelet transform for hierarchical feature capture. Subsequently, we incorporate Manhattan attention to strengthen the detail representation.
3) We propose the HetCoLearn framework with a two-stage collaborative architecture. The framework implements a three-level collaborative learning mechanism, which performs bidirectional KD in feature extraction, accomplishes collaborative weight correction in attention optimization, and enforces hierarchical feature consistency in decoding.

II. Related Work

A. Single-Modal Semantic Segmentation

The past decade witnessed significant advancements in deep-learning-based RGB semantic segmentation techniques. One foundational contribution in this field originates from the research by Long et al. [8], who pioneered the adaptation of mature image classification networks into fully convolutional architectures and achieved end-to-end pixel-level semantic predictions through upsampling operations. Subsequent studies have explored finer grained segmentation methods from multiple perspectives. Zhang et al. [28] proposed the Mamba framework that integrates image registration through object-level structural priors and adaptively aligns multitemporal HSIs by leveraging semantically consistent region masks extracted from pseudo-color projections using the Segment Anything Model. Yao et al. [29] proposed an uncertainty-aggregation-based global–local fusion network (UAGLNet) that effectively leverages high-quality global–local visual semantics guided by uncertainty modeling. He et al. [30] proposed the multiscale aware-relation network (MANet) by exploring multiscale representation methods with distinctiveness and diversity. Yang et al. [31] proposed the channel-position state-space model network (CPSSNet), which adopts an innovative streaming channel modeling method based on state-space models. Liu et al. [32] proposed a novel context aggregation network (CATNet) that aggregates global visual context in the feature, spatial, and instance domains. These advancements have significantly enhanced the capabilities of RGB semantic segmentation models in capturing contextual and structural information.

Despite the advances made in capturing multiscale context and global dependencies, effectively extracting hierarchical and edge-aware features from single-modal RGB data remains a challenge. To overcome this issue, we propose the MWMNet. Our approach uses a wavelet transform to decompose features into multifrequency components, capturing both fine details and coarse structures. In addition, we incorporate a Manhattan attention mechanism for precise spatial and boundary modeling.

B. Multimodal Semantic Segmentation

In recent years, researchers have proposed various innovative methods from the perspective of feature fusion and enhancement to further improve semantic segmentation performance. Zhou et al. [12] proposed the three-stream self-attention network (TSNet), integrating multimodal fusion with attention mechanisms. To incorporate structural information, Xu et al. [33] introduced the high-resolution context extraction network (HRCNet), which leverages edge features to improve

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detail representations. Li et al. [34] proposed a progressive uncertainty-guided segmentation network (PUGNet) for feature fusion optimization, wherein dedicated modules were designed to handle different types of uncertainties in a stage-wise manner. Yang et al. [35] adopted a dynamic weighting strategy to guide fusion ratios and proposed the multilevel gated collaborative network (MGCNet) to effectively reduce cross-modal feature variance. In addition, some researchers employed global semantic enhancement to improve contextual relevance and enhance fusion accuracy. Zhong et al. [36] proposed an attention-based multimodal discrepancy alignment network (AMDANet) that uses low-weight features as cues to alleviate intermodal inconsistency and achieve multimodal feature alignment. Guo et al. [37] proposed a dual-branch CNN architecture (DSNet) that incorporates dilated convolutions into the shallow layers of the model. Cao et al. [38] proposed a CNN- and VMamba-based network with a dual-dimensional differential recalibration mechanism for feature quality enhancement using dimension-specific enhancement strategies to highlight core changes and optimize features. Nevertheless, many of the existing frameworks are limited by their handcrafted fusion architectures, which lack the dynamic adaptability and computational efficiency required for practical applications.

The complex backgrounds in optical remote sensing images pose a significant challenge for accurate semantic segmentation. To address this challenge, various multimodal fusion and lightweight networks have been proposed [39], [40]. Some scholars [41], [42] have explored target detection and semantic segmentation by leveraging the semantic associations of adjacent regions in the context. Lu et al. [43] proposed the context-aware interaction network (CAINet), which explicitly guides learning by constructing an interaction space that uses auxiliary tasks and a global context. These models perform well in multimodal fusion but remain vulnerable to noise from low-quality input images. To address this, several models have been proposed to regulate auxiliary modality fusion and deep information integration [44], [45]. Gao et al. [46] proposed an adaptive frequency enhancement network (AFENet) that adaptively generates separate masks for the high- and low-frequency components, providing optimal details for ground target feature representation.

Building on these advancements, we propose FGMNet to tackle persistent challenges such as noise interference and weak feature fusion in complex remote sensing scenarios. FGMNet, which is designed for RGB and nDSM inputs, introduces two complementary components: a Fourier fusion module that transforms cross-modal features into the frequency domain to separate structural details from noise and enhance fusion effectiveness, and a Gaussian attention mechanism that adaptively adjusts the feature weights based on local reliability.

C. Collaborative Learning

The conventional knowledge collaboration paradigm operates offline, utilizing pretrained large teacher models to guide smaller student models. Foundational work [47] introduced a temperature parameter to soften the output class probability distribution of the teacher. Subsequent research has focused on extracting richer intermediate knowledge representations from the teacher [48], [49]. Wen et al. [50] further enhanced the supervision signal by correcting the teacher's own erroneous predictions. Self-distillation offers a distinct approach in which the student model itself assumes the role of the teacher; for instance, Zhang et al. [51] facilitated knowledge transfer between layers via transformations of internal feature maps. Wang et al. [52] analyzed the characteristics of and assigned customized knowledge schemes and weights to each intermediate layer. Feng et al. [53] embedded edge information as auxiliary supervision during modal fusion and proposed cross-modal edge-privileged KD (CEKD). Although these methods have achieved significant model compression, they share a core limitation: dependence on pretrained teachers. Crucially, the resulting capability gap between the teacher and student ultimately limits further performance gains.

Online knowledge collaboration overcomes the reliance on large pretrained teacher models by enabling the collaborative training of multiple student networks. The core principle of such methods [54] is aligning the softened logit outputs across the student networks to facilitate collaborative learning. For example, Han et al. [55] used asymmetric masks to create discrepancies between different modalities and to adopt a discrepancy-guided complementary learning strategy to achieve mutual learning. In contrast, the relation KD (RKD) method [56] guides a single student model using multilevel information that is extracted from a teacher model. However, RKD relies on a fixed one-way transfer from teacher to student and lacks cross-layer supervision via intermediate features. Thus, the multilevel representations are underutilized.

We propose HetCoLearn, a collaborative framework in which two networks are jointly trained throughout the learning process. Specifically, it involves a collaborative learning mechanism operating concurrently on both networks at three levels: 1) bidirectional knowledge sharing during feature extraction for collaborative guidance; 2) collaborative weight correction within attention optimization; and 3) hierarchical feature consistency enforcement during decoding. This multilevel approach enhances information utilization across network depths and improves overall performance.

Unlike traditional KD, which relies on a fixed one-way transfer from a pretrained teacher to a student, HetCoLearn differs from standard mutual learning frameworks that typically align outputs between homogeneous architectures. While standard mutual learning often employs identical networks (e.g., two ResNets) learning from identical inputs, HetCoLearn bridges the gap between a multimodal frequency-domain network and a unimodal spatial-frequency network. Such heterogeneity enables more effective knowledge exchange. In addition, most online KD methods focus on logit-level alignments. In contrast, our framework implements a three-level collaboration mechanism. By explicitly aligning specialized attention maps and hierarchical decoder features, the framework ensures that the collaborative guidance is consistent across multiple levels of abstraction, rather than being confined to the final output.

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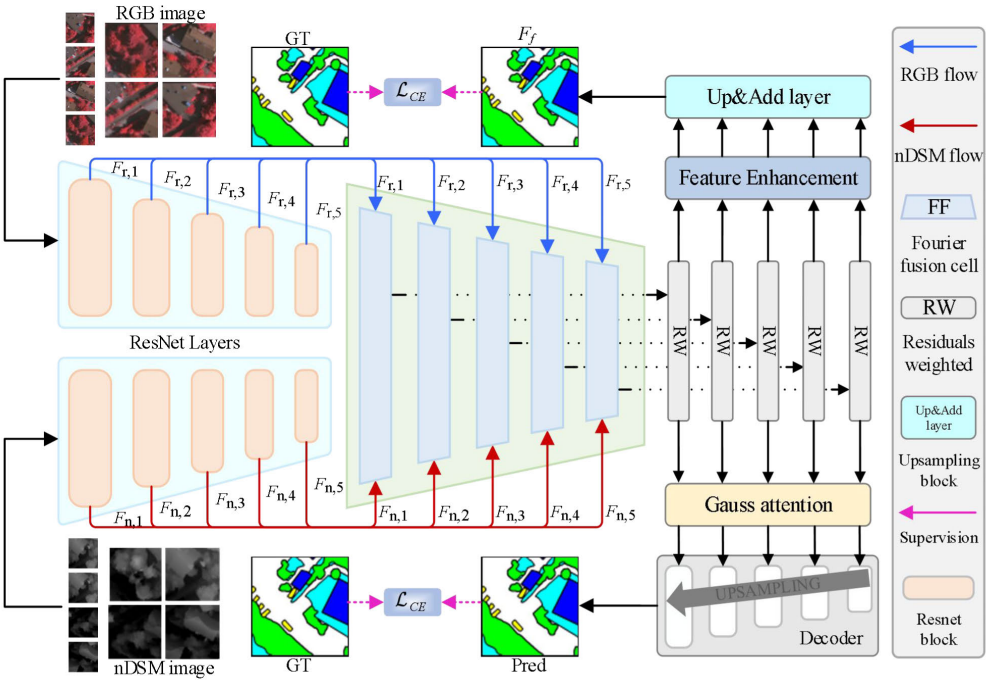


Fig. 1. Our proposed FGMNet.

III. Proposed Network

The proposed methodological framework and its core innovations are described in this section. Specifically, we introduce two novel networks that are designed for distinct data modalities: 1) FGMNet, which simultaneously processes RGB and nDSM inputs and 2) MWMNet, which is specifically designed for single-modal RGB data. Subsequently, we describe the HetCoLearn framework, which connects and co-optimizes these two distinct networks. HetCoLearn incorporates a three-stage collaborative strategy to achieve three-tiered collaborative learning across networks, thereby enhancing the overall performance of the FGMNet and MWMNet.

A. Frequency-Domain Gaussian Multimodal Network

In optical remote sensing, many land cover types exhibit high spectral similarity in the RGB bands (e.g., building roofs versus paved surfaces and trees versus low vegetation), whereas elevation data, such as that provided by nDSM, offer additional geometric clues to distinguish them. As shown in Fig. 1, the proposed FGMNet adopts a dual-branch encoder–decoder architecture to process RGB images and nDSM data. Each modality is first processed in an independent branch to extract unique features without early interference. Next, a Fourier fusion module transforms these features into the frequency domain, aligning their phases and amplitudes to maximize complementary information and reduce redundancy. The merged features are then refined through a residual convolution block, which leverages skip connections to stabilize training and optimize gradient flow. Following attention enhancement, the intermediate features are upsampled to restore spatial resolution under direct training supervision for adaptive recalibration. Finally, a Gaussian attention mechanism applies a Gaussian-weighted map to suppress noise and highlight semantically salient regions, ultimately improving feature quality and model interpretability. Each component is described in detail in Sections II-B and II-C.

1) Encoder: FGMNet employs a standard encoder–decoder structure. Its core, as illustrated in Fig. 1, is a symmetric dual-branch encoder that is designed to independently extract features from RGB and nDSM images. The default five feature extraction layers of the ResNet-18 backbone network are used to process the RGB and nDSM images in parallel, extracting their features to obtain RGB features $F_{r,i}$ (i = 1, 2, 3, 4, and 5) and nDSM features $F_{n,i}$(i = 1, 2, 3, 4, and 5). Thereafter, the five-layer feature maps from the two feature streams are fed into the subsequent fusion module in a one-to-one manner at the same level.

2) Fourier Fusion Cell: Directly fusing RGB images and nDSM data in the spatial domain often aggravates the propagation of local noise and introduces blurring of structural details due to improper modality alignment. To address this, we propose the Fourier fusion cell (FFC), which redistributes cross-modal interactions into the frequency domain. By transforming spatial features into the global frequency domain, it effectively separates structural correlations from high-frequency modality-specific noise, thereby jointly exploiting shared semantic landmarks and distinctive spectral–elevation complementary cues.

A detailed diagram of the FFC is provided in Fig. 2. The RGB features $F_{r,i}$ (i = 1, 2, 3, 4, and 5) and nDSM features $F_{n,i}$ (i = 1, 2, 3, 4, and 5) are first fed into the Fourier fusion block to achieve cross-modal interactions and fusion in the frequency domain. This block processes the inputs in five key stages. Initially, both the RGB and nDSM features undergo a

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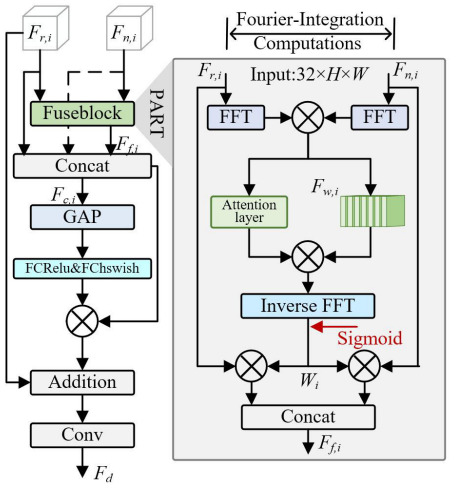


Fig. 2. Structure of the FFC.

2-D fast Fourier transform (FFT), converting them from the spatial domain into complex spectral representations. These bimodal spectra are then multiplied elementwise to produce an initial fused spectrum. Next, frequency–space attention is applied to the fused spectral features to generate attention feature maps. These maps are used to modulate the fused spectrum, which is subsequently transformed back into the spatial domain using an inverse FFT (IFFT). The reconstructed spatial features are then passed through a sigmoid activation function to generate spatial attention weights. Finally, these spatial attention weights modulate the original features $F_{r,i}$ and $F_{n,i}$, which are concatenated along the channel dimension to produce the output

$$F_{w,i} = FFT\left(F_{r,i}\right) \times FFT\left(F_{n,i}\right) \tag{1}$$

$$W_i = Sig\left(IFFT\left(Attention\left(F_{w,i}\right) \times F_{w,i}\right)\right) \tag{2}$$

$$F_{f,i} = Cat\left(\left(F_{r,i} \times W_i\right), \left(F_{n,i} \times W_i\right)\right) \tag{3}$$

where $FFT\ (\cdot)$ and $IFFT\ (\cdot)$ denote the FFT and IFFT functions, respectively, $Sig\ (\cdot)$ and $Attention\ (\cdot)$ denote the sigmoid function and attention weighting layer operation, respectively, and $Cat\ (\cdot)$ denotes the concatenation function.

The fused features are first combined with the original RGB and nDSM inputs into a three-channel composite. This is compressed via global average pooling (GAP) and then processed through a two-layer bottleneck with ReLU and Hardswish activations to generate channel-wise weights. Subsequently, the weighted features are reduced using a 1×1 convolution, and finally, fused with the original RGB data via a residual connection to produce the enhanced output

$$F_{c,i} = Cat\left(F_{r,i}, F_{n,i}, F_{f,i}\right) \tag{4}$$

$$F_d = Conv_{1\times1}\left(FRFH\left(GAP\left(F_{c,i}\right)\right) \times F_{c,i} + F_{r,i}\right) \tag{5}$$

where $GAP\ (\cdot)$ denotes GAP, $FRFH\ (\cdot)$ denotes the operation process incorporating fully connected layers with ReLU and Hardswish activation functions, and $Conv_{1\times1}\ (\cdot)$ denotes the 1×1 convolution operation.

The fused features are then fed into a residual weighting module that integrates asymmetric convolutions to deeply refine their spatial details and channel weights. This module combines multilevel feature extraction with a dual attention mechanism, capturing features through two cascaded stages. In the first stage, dilated convolution (dilation = 3) and asymmetric convolutions (1×3 and 3×1) are used in parallel to balance the receptive field and for computational efficiency. In the second stage, the dilation rate is increased to 5, accompanied by wider asymmetric convolutions (1×5 and 5×1) to capture more global semantic information. After each stage, feature concatenation is performed, and the current features are accumulated with the historical inputs via multilevel skip connections, thereby ensuring efficient information flow.

3) Adaptive Gaussian Attention Focus Module: In remote sensing image processing, nDSM data are often corrupted by sensor noise, intraobject irregularities, and void structures. Contrary to traditional spatial attention mechanisms, which typically rely on pixel-wise convolutional mappings and are highly prone to amplifying local boundary noise and void interference, we propose an adaptive Gaussian focusing (AGF) module. By imposing a global spatial smoothing constraint through a 2-D prior distribution, the AGF module effectively filters out high-frequency boundary noise fluctuations and focuses attention precisely on the stable core regions of objects. For a given input feature map, the module models the mean and log variance of a Gaussian distribution as learnable parameters, enabling adaptive adjustment of the attention center and coverage range during training. Specifically, the process first computes the 1-D Gaussian vectors along the height and width dimensions in a normalized coordinate space. An outer product operation then combines them to construct a 2-D Gaussian attention map. Finally, this map is normalized by a Sigmoid function and multiplied element-wise with the original feature map. The weighted features are then added back to the original input through a residual connection. Mathematically, this process is expressed as follows:

$$\begin{aligned} y_i &= \frac{i}{H-1}, \quad x_j = \frac{j}{W-1} \\ i &= 0, \ldots, H-1; \quad j = 0, \ldots, W-1 \end{aligned} \tag{6}$$

$$g_{h,i} = \exp\left(-\frac{(y_i - \mu_{\mathrm{h}})^2}{2\mathrm{e}^{\mathrm{hlog}} + 10^{-6}}\right)$$

$$g_{w,j} = \exp\left(-\frac{\left(x_j - \mu_{\mathrm{w}}\right)^2}{2\mathrm{e}^{\mathrm{wlog}} + 10^{-6}}\right) \tag{7}$$

$$G_{i,j} = g_{h,i} \times g_{w,j} \tag{8}$$

where μ_h and μ_w denote the mean values of the Gaussian distribution in the height and width directions, respectively, $h$log and $w$log denote the logarithmic variances of the Gaussian distribution in the respective directions, $g_{h,i}$ and $g_{h,j}$ represent 1-D Gaussian weights, and $G_{i,j}$ denotes the 2-D weight matrix attention. Subsequently, the generated attention features are expanded back into four dimensions and weighted to the original input features. Finally, residual connections are applied to supplement the lost information.

4) Decoder: Before the final supervised processing is applied to the features, the acquired multilevel features first undergo feature enhancement. Specifically, the input feature

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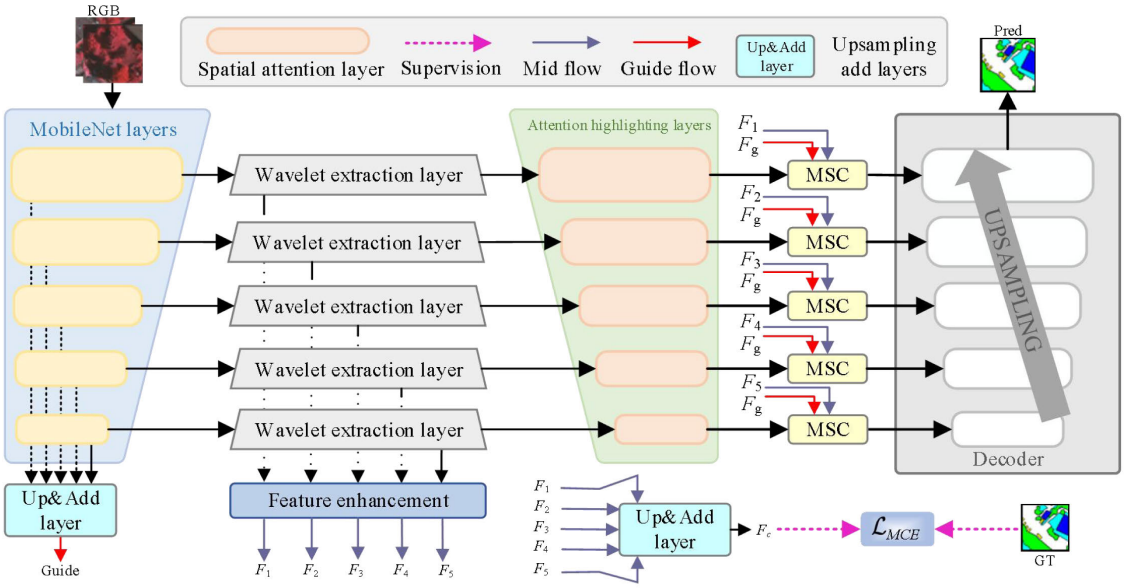


Fig. 3. Proposed MWMNet.

map is processed by a channel attention module to obtain weighted features along the channel dimensions. The output is then passed to a spatial attention module, which generates a spatial weight map. Finally, the channel-attention output and the spatial weight map are fused via element-wise multiplication, producing a feature map that emphasizes the important channels and key spatial regions. Our decoder employs a progressive fusion strategy that integrates features from the deeper layers into shallower ones. The higher level features are upsampled to match the resolution of the lower level features and then combined through elementwise multiplication to enhance the semantic information and highlight target-relevant features. The refined features are further processed using convolution and residual connections, preserving the spatial details while adding a semantic context. Finally, the adjacent layer features are concatenated and fused via channel compression convolution.

During training, the model not only computes the cross-entropy (CE) loss on the final predicted feature map (Pred) but also introduces an auxiliary supervision branch, where the features F_{mid} extracted from an intermediate frequency-domain layer are used to compute the CE loss against the ground truth (GT). Both losses share the same pixel-level GT labels, enabling the simultaneous optimization of the intermediate frequency-domain features and the final prediction.

B. MWMNet

We propose MWMNet, which integrates a multiscale wavelet transform and Manhattan attention to address the limitations of single-modal RGB data in multiscale feature representation and edge extraction. As illustrated in Fig. 3, the network follows an encoder–decoder framework. First, a multiscale wavelet transform module decomposes the input images into frequency components to construct enriched hierarchical representations. Wavelet functions provide a more effective time–frequency representation for capturing local texture and edge features compared with traditional convolutional feature extraction methods. Next, the Manhattan attention mechanism enhances the geometric integrity of the object boundaries by emphasizing critical contour regions. These modules are integrated into a symmetric encoder–decoder network to progressively restore spatial resolution while preserving high-frequency details. The following sections provide detailed descriptions of the network architecture and individual component designs.

1) Encoder: The MWMNet employs a standard encoder–decoder architecture. As illustrated in Fig. 3, the core component is a multiscale feature extraction encoder that is designed to capture hierarchical feature representations from the input image. The features are first extracted using a MobileNetV2 backbone network, generating five feature maps (denoted as F_1–F_5) with distinct spatial resolutions and channel dimensions of $16 \times 128 \times 128$, $24 \times 64 \times 64$, $32 \times 32 \times 32$, $160 \times 16 \times 16$, and $320 \times 8 \times 8$. Among them, F_1 is extracted from the first two layers of MobileNetV2 and corresponds to shallow texture features; F_2 is extracted from layers 2–4, F_3 from layers 4–7, F_4 from layers 7–17, and F_5 from the last layer, thereby forming a multilevel feature pyramid ranging from high-resolution low-semantic features to low-resolution high-semantic features. These feature maps first pass through selective kernel attention [57] to mitigate the information loss that may be caused by subsequent reduction in dimensions. They are then uniformly reduced to 32 channels using a dedicated dimension-reduction convolution module. Finally, the reduced feature maps are fed into the subsequent multimodal fusion-decoding module in a layer-wise manner.

2) Wavelet Extraction Layer: Traditional methods often face challenges such as local noise and insufficient multiscale feature capture in high-resolution remote sensing image

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segmentation. Therefore, we introduce the wavelet extraction layer (WEL), which uses parallel wavelet transforms and convolutional attention branches to refine and fuse multiscale features.

Specifically, the input feature map is subjected to multiscale extraction and fusion using a wavelet transform module with three parallel branches. Each branch decomposes and reconstructs features using wavelet transforms while simultaneously applying channel and spatial attention to weight the inputs adaptively. The weighted features are then processed using an asymmetric convolution block. The outputs from both paths are concatenated channel-wise and fused via a 3×3 convolution. Finally, the results from all three branches are summed, adjusted using a 1×1 convolution, and multiplied elementwise with dilated (rate = 7) 3×3 convolutional features from the original input. Residual connections integrate the original information to produce the final output

$$ConvL_j = Conv_{1\times1}\left(Conv_{1\times(2j+1)}\left(Conv_{(2j+1)\times1}(\cdot)\right)\right) \quad j = 1,2,3 \tag{9}$$

$$F_{c,i}^{j} = ConvL_j\left(Sa\left(Ca\left(F_i\right)\times F_i\right)\times F_i\right), \quad i = 1,\ldots,5 \tag{10}$$

$$F_{o,i}^{1} = Conv\left(Cat\left(WLC_1\left(F_i\right), F_{c,i}^{1}\right)\right) \tag{11}$$

$$F_{o,i}^{2} = Conv\left(Cat\left(WLC_2\left(F_i\right), F_{c,i}^{2}\right)\right) \tag{12}$$

$$F_{o,i}^{3} = Conv\left(Cat\left(WLC_3\left(F_i\right), F_{c,i}^{3}\right)\right) \tag{13}$$

$$O_i = Mul\left(Conv\left(F_{o,i}^{1} + F_{o,i}^{2} + F_{o,i}^{3}\right), Conv\left(F_i\right)\right) + F_i \tag{14}$$

where $ConvL(\cdot)$ denotes the asymmetric convolution block layer, $Ca(\cdot)$ denotes the channel attention operation, $Sa(\cdot)$ denotes the spatial attention operation, $WLC(\cdot)$ denotes the wavelet transform convolution block, $Conv(\cdot)$ denotes the convolution operation, $Mul(\cdot)$ denotes the multiply function, F_i denotes the original input feature, $F_{c,i}$ denotes the attention-enhanced feature, $F_{o,i}$ denotes the output feature of each wavelet convolution branch, and O_i denotes the final output feature of the WEL.

The wavelet convolution block employs the Haar wavelet as the base function to perform discrete wavelet transform (DWT), which decomposes the input features into frequency-domain representations. Specifically, the DWT breaks the features down into four subbands: one low-frequency approximation (LL) and three high-frequency details (LH, HL, and HH) representing horizontal, vertical, and diagonal orientations, respectively. These subbands are concatenated along the channel dimension to preserve multifrequency information. These subbands are concatenated along the channel dimensions and fused using a 3×3 dilated convolution (dilation rate = 3), followed by batch normalization and ReLU processing. The enhanced features are then decomposed again into four subbands and reconstructed via an inverse DWT, maintaining the original input resolution while effectively integrating the multifrequency information

$$WCL = IDWT\left(ReLU\left(BN\left(Conv_{3\times3}\left(d=3, DWT\left(\cdot\right)\right)\right)\right)\right) \tag{15}$$

where $DWT(\cdot)$ denotes the DWT, $IDWT(\cdot)$ denotes the inverse DWT, $Conv_{3\times3}(\cdot)$ represents a convolution operation with a 3×3 kernel and a dilation rate of 3, and $BN(\cdot)$ denotes the batch normalization function.

The extracted features then enter a hybrid attention layer for adaptive recalibration. First, the features are weighted by channel attention and combined with a residual connection to produce channel-enhanced features. Subsequently, spatial attention weights are generated from these features to perform element-wise recalibration on the original input. Finally, residual mapping is introduced once more.

3) Manhattan Synergy Core: In urban remote sensing imagery, where buildings and roads typically form regular, rectangular, or grid-like patterns, Manhattan attention [58] offers greater sensitivity to axial structural variations than traditional Euclidean distance or dot-product attention, enabling more accurate delineation of linear ground features. Single-modal features often fail to capture spatial relationships and global context in complex scenes. We propose adopting the Manhattan synergy core (MSC) to overcome this problem.

As shown in Fig. 3, the Manhattan attention module uses the feature map F_i, Manhattan branch feature F_m, and global branch feature F_g as inputs to enable cross-branch feature enhancement and fusion. The module first extracts spatial relationships through Manhattan position encoding and captures long-range dependencies using the Manhattan attention mechanism. Concurrently, it generates a spatial weight map via feature branching to reweight and update F_m and F_g adaptively. The updated branch features are then concatenated along the channel dimensions and fused through a 1×1 convolution

$$A_i = Manhattan\left(F_i\right), \quad i = 1,\ldots,5 \tag{16}$$

$$S_i = Sig\left(Conv\left(ReLU\left(BN\left(Conv\left(GAP\left(A_i\right)\right)\right)\right)\right)\right) \tag{17}$$

$$G_{m,i} = Conv\left(S_i \times F_{m,i}\right) + F_{m,i} \tag{18}$$

$$G_g = Conv\left(S_i \times F_g\right) + F_g \tag{19}$$

$$G_{f,i} = Conv\left(Cat\left(G_{m,i}, G_g\right)\right) \tag{20}$$

where $Manhattan(\cdot)$ denotes the Manhattan self-attention block, A_i denotes the Manhattan enhanced feature, S_i denotes the Manhattan attention feature map, and $G_{f,i}$ denotes the multiscale enhanced fusion feature. Simultaneously, the channel and spatial attention mechanisms weight and modulate the Manhattan attention features. Finally, the fused-branch features are combined with these attention-modulated features, refined via a 1×1 convolution, and integrated with the original input F_i through residual connections to produce the final output

$$G_{a,i} = G_{f,i} + Sa\left(Ca\left(A_i\right)\right) \tag{21}$$

$$O_i = Conv\left(G_{a,i}\right) + F_i \tag{22}$$

where O_i denotes the final enhanced feature output by the Manhattan collaborative core module.

4) Decoder: As this decoder uses the same layered decoder architecture as FGMNet, we have not detailed this process here.

C. Heterogeneous Collaborative Learning

As illustrated in Fig. 4, the HetCoLearn framework adopts a three-level feature interaction mechanism for parallel optimization during training.

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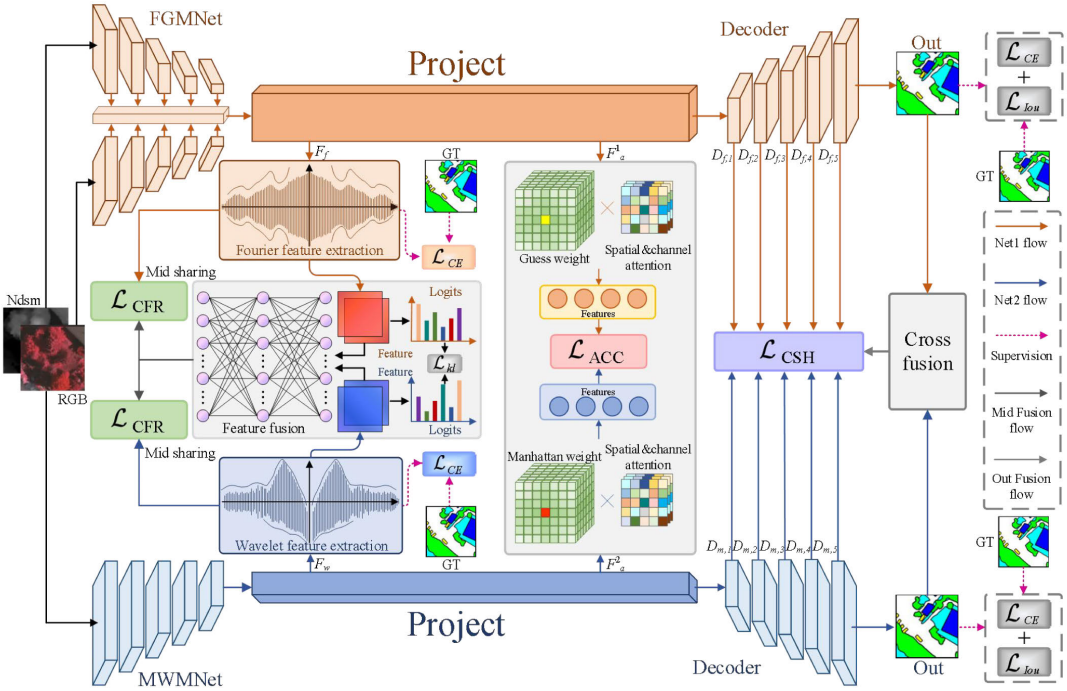


Fig. 4. Proposed HetCoLearn.

1) Collaborative Feature Refinement: Global frequency-domain correlations and local spectral details are inherently complementary. The FGMNet incorporates a Fourier fusion unit that transforms cross-modal features into a complex spectral field via the two-dimensional FFT. This enables the model to align cross-modal phase and amplitude patterns from a global frequency perspective, thereby capturing macrolevel scene semantics. MWMNet, on the contrary, relies on a wavelet extraction layer that employs the DWT to decompose spatial features into multiple local frequency subbands. This endows the MWMNet with superior local time–frequency localization, allowing it to focus intensely on microlevel information variations—details that are often smoothed out by global Fourier operations.

The FGMNet effectively captures globally coherent structures but fails to preserve fine-grained local details. In contrast, the MWMNet decomposes the multiresolution, high-frequency directional details through the WEL; however, it is entirely devoid of the geometric elevation cues provided by the nDSM. By implementing bidirectional KD at this stage, collaborative feature refinement (CFR) enables the MWMNet to implicitly absorb elevation structural knowledge and allows the FGMNet to inherit fine texture information.

During this stage, we use the classic Kullback–Leibler (KL) loss function to match pairs and improve learning effectiveness. The standard KL loss function [20] can be expressed as

$$KL\left(Y^1 \| Y^2\right) = \frac{\tau^2}{B} \sum_{b=1}^{B} \sum_{c=1}^{C} Y_{b,c}^1 \log\left(\frac{Y_{b,c}^1}{Y_{b,c}^2}\right) \tag{23}$$

where KL is the KL divergence; B and C denote the batch size and number of classes, respectively; and τ is the control temperature. The CE loss function is used during the supervision of individual frequency-domain features

$$CE = -\frac{1}{W \times H} \sum_{i=1}^{W} \sum_{j=1}^{H} G_{i,j} \log\left(P_{i,j}\right) + \left(1 - G_{i,j}\right) \log\left(1 - P_{i,j}\right) \tag{24}$$

where $G_{i,j}$ and $P_{i,j}$ denote the pixel values of the GT and prediction result at (i, j) within the image, respectively, and H and W denote the height and width of the computed image, respectively.

The frequency-domain features F_f extracted by the Fourier module in the FGMNet and F_w extracted by the wavelet module in the MWMNet are calculated using the KL loss function to determine the semantic association. The two features are then merged through lightweight gated fusion. Guidance signals are separately extracted from the two feature streams, and a lightweight convolution followed by a Sigmoid function dynamically generates channel-wise gating weights, which are used to perform a complementary weighted summation of the two feature streams. The KL loss is calculated for the original and merged features. In addition, these two frequency-domain features are supervised separately using the CE loss function

$$L_{sup}^1 = CE\left(F_f, GT\right) + CE\left(F_w, GT\right) \tag{25}$$

$$L_{KL} = KL\left(F_f, F_w\right) + KL\left(F_{fuse}, F_f\right) + KL\left(F_{fuse}, F_w\right). \tag{26}$$

The overall loss can be expressed as

$$L_{CFR} = L_{sup}^1 + L_{KL}. \tag{27}$$

2) ACC Mechanism: The isotropically smoothed features are complemented by the anisotropic axial features. The FGMNet introduces a Gaussian attention focusing module,

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which imposes an isotropic spatial smoothing constraint via a 2-D Gaussian prior distribution to filter out high-frequency spatial noise fluctuations, thereby anchoring the core semantic masks of large objects. In contrast, the MWMNet adopts a Manhattan attention kernel. Because urban remote sensing scenes predominantly exhibit orthogonal and axial structures, the Manhattan distance metric is particularly sensitive to linear structural variations.

The Gaussian attention of the FGMNet excels at suppressing high-frequency noise fluctuations to ensure intraregion consistency, whereas the MWMNet's Manhattan attention is more sensitive to axially regular targets such as elevated points and roads. Through weight calibration and mutual guidance between the two attention modules, the attention collaborative correction (ACC) mechanism achieves dual optimization, namely noise elimination and high-fidelity detail preservation, thereby resolving the incompatible tradeoff between global regional feature stability and local target detail precision in complex remote sensing scenes.

The KD loss function is still used for knowledge matching and sharing. The attention features F_{a}^{1} after the Gaussian attention in the FGMNet and F_{a}^{2} after the Manhattan attention in MWMNet share attention information through the KD loss

$$L_{AMC} = KL\left(F_{a}^{1}, F_{a}^{2}\right). \tag{28}$$

In summary, this stage of the collaborative learning achieves synergy between the two attention guidance strategies, thereby enhancing the ability of the model to perform refined feature selection and spatial relationship modeling in complex scenarios.

3) CSH Feature Learning: As features enter the decoder, the predictions of the two networks tend to produce inconsistent optimization directions at complex edges or for long-tail categories. The collaboratively supervised hierarchical (CSH) stage imposes layer-wise contrastive constraints on the corresponding decoder levels of both networks, ensuring that the heterogeneous branches reach semantic consensus at the same abstraction level.

We use the cosine loss function to perform cross-network comparisons at the same level

$$\mathrm{Cos} = \frac{1}{N}\sum_{i=1}^{N}\left(1 - \cos\left(\theta_i\right)\right) \tag{29}$$

where $N(\cdot)$ denotes the batch size and $\theta(\cdot)$ denotes the angle between the input features. Specifically, the hierarchical features $D_{w,i}$ in the decoder part of FGMNet and $D_{m,i}$ in the decoder part of MWMNet are compared layer by layer. In addition, the final output features of the two networks are merged, and the original output features are compared for loss learning

$$L_{\mathrm{COS}}^{1} = \sum_{i=1}^{5}\mathrm{Cos}\left(D_{f,i}, D_{m,i}\right) \tag{30}$$

$$L_{\mathrm{COS}}^{2} = \mathrm{Cos}\left(D_{f,5}, D_{fuse}\right) + \mathrm{Cos}\left(D_{m,5}, D_{fuse}\right) \tag{31}$$

$$L_{MSH} = L_{\mathrm{COS}}^{1} + L_{\mathrm{COS}}^{2}. \tag{32}$$

This component of contrastive learning ensures consistent understanding across different networks at the same semantic level, improving semantic gaps and information degradation in the hierarchical feature fusion and providing a multilevel feature foundation for high-precision predictions.

4) Total Loss: At the end of the framework, the final outputs of the two networks are supervised using the CE and intersection over union (IoU) losses

$$L_{out}^{1} = CE\left(D_{f,5}, GT\right) + IOU\left(D_{f,5}, GT\right) \tag{33}$$

$$L_{out}^{2} = CE\left(D_{m,5}, GT\right) + IOU\left(D_{m,5}, GT\right) \tag{34}$$

$$L_{out} = L_{out}^{1} + L_{out}^{2}. \tag{35}$$

The IoU loss considers the overall spatial overlap between the GT graph G and prediction map S, rather than individual pixels, forcing the model to focus more on defective pixels during training. The IoU loss is formulated as

$$IOU = 1 - \frac{\sum_{i=1}^{H}\sum_{j=1}^{W} G_{i,j}\cdot S_{i,j}}{\sum_{i=1}^{H}\sum_{j=1}^{W}\left(G_{i,j} + S_{i,j} - G_{i,j}\cdot S_{i,j}\right)} \tag{36}$$

where H and W denote the height and width of G, respectively, and $G_{i,j}$ and $S_{i,j}$ denote the label and prediction of G and S at pixel (i, j), respectively.

The total loss is obtained after organizing the losses from the three mutual learning stages. The supervision loss and all loss weight coefficients are set to 1 by default. The overall loss formula is as follows:

$$L_{total} = L_{CFR} + L_{AMC} + L_{MSH} + L_{\mathrm{out}}. \tag{37}$$

IV. Experiments and Results

To evaluate the effectiveness of the proposed model, we conducted a comprehensive comparison with 16 state-of-the-art models across two categories—single-modal networks and multimodal methods. Single-modal networks include classical models, such as FCN-8 S [8], U-Net [9], SegNet [10], and the more advanced CATNet [32], UAGLNet [29], HRCNet [33] AFENet [46], and multimodal methods include TSNet [12], MGCNet [35], MANet [30], DCSIUNet [35], DSNet [37], AMDANet [36], PUGNet [34], CEKD [54], CAINet [43], and CPSSNet [31].

A. Dataset and Metrics

1) Datasets: The proposed network was validated using two widely used public datasets for remote sensing image segmentation: Vaihingen and Potsdam. Both datasets comprise high-resolution aerial images and include annotations for six common land-cover classes: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background. These datasets serve as standard benchmarks for evaluating semantic segmentation models in the remote sensing field.

2) Metrics: To evaluate the classification performance comprehensively, we employed four standard metrics: the class-wise IoU, class-wise accuracy (Acc), and their global averages—the mean IoU (mIoU) and mean accuracy (mAcc), respectively.

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TABLE I

QUANTITATIVE EVALUATION OF MODELS ON THE VAIHINGEN DATASET

	Imp.surf		Tree		Low veg.		Building		Car		mAcc	mIoU
	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
U-Net$_{2015}$ [9]	91.68	80.90	91.30	77.86	77.97	65.91	89.84	86.50	75.80	71.22	79.75	71.34
FCN-8S$_{2015}$ [8]	89.66	79.71	89.22	75.58	75.83	64.33	93.22	86.80	45.12	40.16	78.61	69.32
SegNet$_{2017}$ [10]	89.88	80.93	88.96	75.96	78.66	64.07	90.88	86.54	43.93	43.16	78.46	70.13
MANet$_{2022}$ [30]	91.57	83.23	91.91	77.94	76.29	65.59	93.77	90.04	86.03	81.01	87.91	79.56
HRCNet$_{2021}$ [32]	91.62	81.60	90.55	78.58	79.24	67.38	91.72	88.01	70.69	68.73	84.76	76.86
TSNet$_{2020}$ [12]	87.93	78.98	94.26	81.26	71.62	57.03	95.81	91.47	67.63	66.86	83.54	75.12
MGCNet$_{2022}$ [34]	91.18	84.89	92.76	78.53	76.22	65.74	95.58	91.52	79.56	75.33	87.06	79.20
CEKD$_{2023}$ [54]	92.22	85.05	92.69	78.97	76.57	66.77	95.78	91.66	76.63	72.01	86.71	78.89
CAINet$_{2024}$ [43]	90.50	83.10	91.16	77.76	77.28	64.94	95.16	91.22	54.11	51.07	81.64	73.62
CPSSNet$_{2025}$ [31]	93.28	83.28	94.02	79.04	71.28	64.44	95.31	90.71	83.05	77.26	87.39	78.95
AFENet$_{2025}$ [46]	91.88	84.52	91.03	78.59	80.29	67.66	93.96	91.02	82.94	78.92	88.02	80.14
DSNet$_{2025}$ [37]	89.53	81.20	91.56	76.75	75.84	64.60	92.38	87.06	68.85	65.69	83.63	75.06
AMDANet$_{2025}$ [36]	90.45	82.79	90.92	77.07	76.94	64.55	94.63	90.48	64.20	62.12	83.43	75.40
CATNet$_{2025}$ [32]	92.72	83.88	91.51	78.60	77.25	66.49	94.39	91.39	86.19	82.12	88.41	80.49
UAGLNet$_{2026}$ [29]	91.01	83.52	91.68	78.23	76.75	66.26	94.63	89.32	80.37	75.50	86.89	78.56
PUGNet$_{2026}$ [33]	91.39	84.40	92.27	78.68	77.33	66.63	95.54	91.53	84.30	80.08	88.17	80.26
FGMNet	90.13	84.81	86.14	78.04	82.46	67.98	93.86	90.62	83.17	77.69	87.15	79.83
MWMNet	90.51	84.72	90.21	77.86	76.32	65.19	94.24	90.39	83.52	77.79	86.96	79.19
FGMNet-CL	91.06	85.08	87.07	78.70	85.57	68.43	94.94	92.09	86.49	81.56	89.03	81.17
MWMNet-CL	91.75	85.24	91.73	78.62	78.60	67.24	95.35	91.84	87.14	80.15	88.91	80.62

TABLE II

QUANTITATIVE EVALUATION OF MODELS ON THE POTSDAM DATASET

	Imp.surf		Tree		Low veg.		Building		Car		Cluter		mAcc	mIoU
	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
U-Net$_{2015}$ [9]	90.03	80.27	84.06	72.05	85.82	71.60	88.71	82.92	93.89	90.24	50.30	36.26	82.13	72.22
FCN-8S$_{2015}$ [8]	89.47	79.77	82.86	71.23	85.13	71.12	90.69	83.60	91.02	81.53	49.05	36.49	78.61	69.32
SegNet$_{2017}$ [10]	90.18	80.46	82.49	70.68	85.88	71.63	90.21	84.18	93.15	89.72	51.76	37.21	82.28	72.31
MANet$_{2022}$ [30]	90.48	81.20	82.69	71.54	88.78	72.74	87.58	83.47	90.96	88.73	46.16	38.11	81.11	72.63
HRCNet$_{2021}$ [32]	90.03	81.68	82.02	71.32	88.17	73.18	90.87	85.75	93.94	89.82	56.72	40.03	83.63	73.63
TSNet$_{2020}$ [12]	85.22	76.85	78.75	67.49	88.52	67.98	91.85	86.65	78.22	76.85	37.49	30.85	76.68	67.78
MGCNet$_{2022}$ [34]	90.61	81.21	83.70	72.51	86.72	72.67	93.87	89.02	93.39	88.91	53.24	39.66	83.59	74.00
CEKD$_{2023}$ [54]	91.09	80.89	85.18	71.62	84.70	71.71	93.37	88.97	93.01	89.35	52.63	38.18	83.33	73.45
CAINet$_{2024}$ [43]	90.52	80.35	81.15	67.89	84.70	70.55	93.13	89.32	93.48	85.33	48.06	37.04	81.84	71.75
CPSSNet$_{2025}$ [31]	88.87	81.23	81.38	71.95	89.02	73.18	91.82	85.61	93.23	89.56	54.76	38.48	83.18	73.34
AFENet$_{2025}$ [46]	91.52	82.18	83.87	72.97	87.26	73.63	90.93	86.52	94.53	90.89	56.84	42.08	84.16	74.71
DSNet$_{2025}$ [37]	90.25	79.55	78.32	66.43	85.72	70.01	87.86	82.52	92.99	88.35	51.54	40.13	81.11	71.16
AMDANet$_{2025}$ [36]	90.27	78.72	79.84	65.68	84.60	68.93	92.21	88.30	91.11	80.45	45.19	35.58	80.54	69.61
CATNet$_{2025}$ [32]	90.61	81.79	84.82	72.46	86.77	72.89	92.03	87.04	94.64	89.46	51.86	40.49	83.45	74.02
UAGLNet$_{2026}$ [29]	90.03	82.16	83.92	71.62	87.84	72.62	90.13	85.64	94.72	88.64	50.72	39.58	82.89	73.38
PUGNet$_{2026}$ [33]	92.12	82.31	85.53	73.08	86.90	72.78	90.88	86.19	94.16	89.84	55.53	40.36	84.19	74.09
FGMNet	90.13	80.32	83.39	71.76	83.22	71.59	91.84	87.57	91.27	87.72	57.44	38.92	82.89	72.98
MWMNet	89.27	80.19	82.88	71.21	83.29	71.18	89.42	86.39	92.34	87.61	58.02	39.14	82.53	72.62
FGMNet-CL	91.60	82.26	85.56	73.72	85.41	73.23	93.17	89.18	93.49	89.94	59.05	40.86	84.71	74.86
MWMNet-CL	90.76	81.77	84.42	72.34	85.57	72.86	90.91	87.53	94.20	89.88	60.48	40.31	84.39	74.12

B. Experimental Setup

We conducted our experiments using PyTorch 1.13.1 and Cuda 11.0, and performed all neural network training on a single NVIDIA TiTanV machine equipped with 12 GB of RAM. We employed the Adam optimizer with a momentum of 0.9, an initial learning rate of 0.0001, and a weight decay of 0.0005 for optimization. The momentum parameter decayed by a factor of 0.1 every 20 cycles.

C. Performance Evaluation

All methods were evaluated on the exact same data splits of the Vaihingen and Potsdam datasets under a unified experimental framework. Unimodal networks utilized only RGB images, whereas multimodal networks were adapted to leverage RGB images combined with the corresponding nDSM data as an auxiliary modality. Furthermore, all models were assessed using identical evaluation scripts and standard metrics, without any test-time augmentation or postprocessing. The relevant data are presented in Tables I and II.

1) Quantitative Results and Analysis: The performances of the proposed method and existing deep-learning models were validated on the Vaihingen and Potsdam datasets through comparative experiments, which were repeated five times to ensure reliability. We denote the two networks after collaborative learning as FGMNet-CL and MWMNet-CL. As shown in Tables I and II, the proposed method achieved competitive performance on both datasets. Following collaborative learning, FGMNet achieved 89.03% mAcc and 81.17% mIoU on the Vaihingen dataset and 84.71% mAcc and 74.86% mIoU on the Potsdam dataset, whereas MWMNet achieved 88.91% mAcc and 80.62% mIoU on the Vaihingen dataset and 84.39% mAcc and 74.12% mIoU on the Potsdam dataset.

2) Visual Comparison and Analysis: Fig. 5 illustrates the advantages of our network using representative examples from

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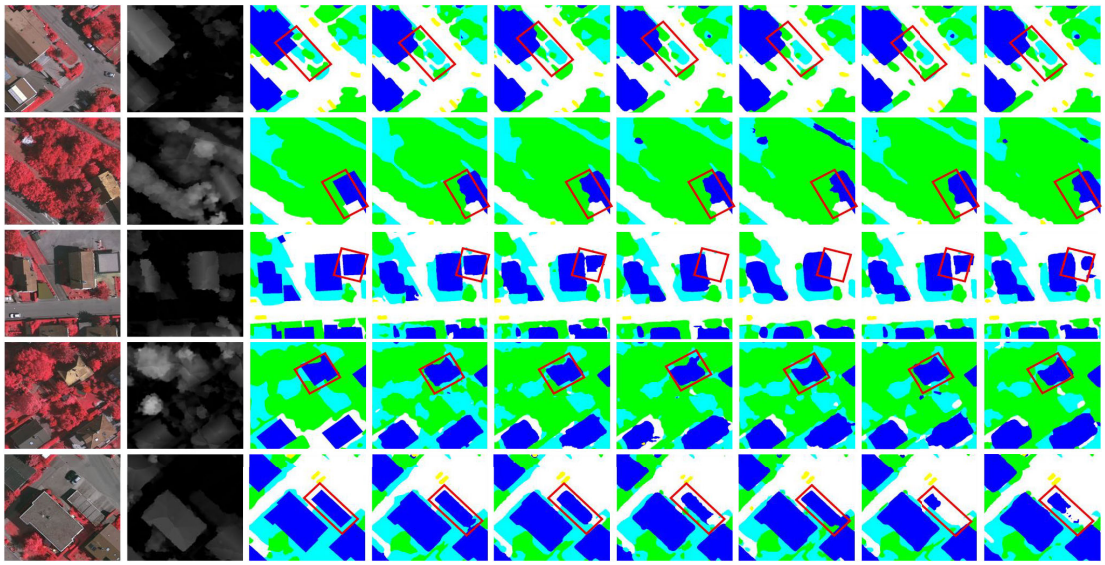


Fig. 5. Semantic segmentation results: (a) RGB, (b) nDSM, (c) GT, (d) FGMNet-CL, (e) MWMNet-CL, (f) DSNet, (g) CEKDNet, (h) PUGNet, and (i) UAGLNet.

TABLE III
COMPUTATIONAL COMPLEXITY COMPARISON

Model	Params (M)↓	FlOPs↓	FPS↑
FCN-8S$_{2015}$ [8]	26.36	55.93	8.7
U-Net$_{2015}$ [9]	134.29	74.55	12.3
HRCNet$_{2021}$ [32]	62.71	30.27	14.6
MGCNet$_{2022}$ [34]	209.99	47.51	9.4
CEKD$_{2023}$ [54]	245.03	68.61	7.2
MAGNet$_{2024}$ [37]	16.1	9.9	31.3
CPSSNet$_{2025}$ [31]	31.86	8.25	7.7
DSNet$_{2025}$ [37]	30.07	33.49	36.49
PUGNet$_{2026}$ [33]	27.55	8.37	22.79
FGMNet-CL	26.75	9.64	29.4
MWMNet-CL	18.24	8.99	20.7
HetCoLearn	45.02	18.85	16.6

the Vaihingen and Potsdam datasets. Compared with five state-of-the-art models, our approach achieved more accurate edge recognition and higher structural integrity for vehicles and buildings, while also capturing finer details of vegetation without large-scale misclassification. These improvements can be attributed to the collaborative learning mechanism, which enables the two heterogeneous networks to exchange complementary information effectively.

3) Complexity Comparison: Our collaborative framework achieved an optimal balance between performance and efficiency (Table III). It delivered higher accuracy than many larger models while maintaining a compact size. The proposed framework enabled efficient operation on limited hardware, making it highly suitable for practical industrial applications.

D. Ablation Experiments

We systematically evaluated the contribution of each network component to the overall performance through comprehensive ablation studies, wherein we progressively incorporated or removed specific modules to analyze their specific impact. The results are presented in Tables IV–VIII, where √ indicates retained components. Tables IV and V present the results of module ablation experiments conducted at the individual submodel training level, whereas Tables VI and VII present for the ablation experiments on the learning strategies for corresponding modules during the collaborative training of the two submodels.

TABLE IV
ABLATION EXPERIMENTS FOR EACH COLLABORATIVE LEARNING MODULE OF THE PROPOSED NETWORK

Methods	FFC	AGF	VAIHINGEN		POTSDAM	
			mAcc	mIoU	mAcc	mIoU
Wo/FFC		√	86.18	78.79	81.63	71.65
Wo/AGF	√		86.03	78.65	81.47	71.79
FGMNet	√	√	87.15	79.83	82.89	72.98

TABLE V
ABLATION EXPERIMENTS FOR EACH MODULE OF THE PROPOSED NETWORK

Methods	WEL	MSC	VAIHINGEN		POTSDAM	
			mAcc	mIoU	mAcc	mIoU
Wo/WEL		√	85.87	78.06	81.42	71.67
Wo/MSC	√		85.91	78.21	81.59	71.74
MWMNet	√	√	86.96	79.19	82.53	72.62

TABLE VI
ABLATION EXPERIMENTS FOR EACH COLLABORATIVE LEARNING MODULE OF OUR PROPOSED NETWORK (FGMNET-CL)

Methods	CFR	ACC	CSH	VAIHINGEN		POTSDAM	
				mAcc	mIoU	mAcc	mIoU
Wo/CFR		√	√	88.06	80.12	83.74	73.84
Wo/ACC	√		√	88.15	80.21	83.69	73.79
Wo/CSH	√	√		87.98	80.04	83.48	73.72
FGMNet-CL	√	√	√	89.03	81.17	84.71	74.86

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TABLE VII

ABLATION EXPERIMENTS FOR EACH COLLABORATIVE LEARNING MODULE OF OUR PROPOSED NETWORK (MWMNET-CL)

Methods	CFR	ACC	CSH	VAIHINGEN		POTSDAM	
				mAcc	mIoU	mAcc	mIoU
Wo/CFR		√	√	87.89	79.49	83.19	73.17
Wo/ACC	√		√	87.96	79.68	83.32	73.15
Wo/CSH	√	√		87.74	79.42	83.17	73.06
MWMNet-CL	√	√	√	88.91	80.62	84.39	74.12

TABLE VIII

ABLATION EXPERIMENTS ON COLLABORATIVE LEARNING TO REMOVE THE NDSM MODALITY

Methods	VAIHINGEN		POTSDAM	
	mAcc	mIoU	mAcc	mIoU
FGMNet-CL	88.53	80.41	84.15	74.23
MWMNet-CL	88.49	80.28	84.04	73.87

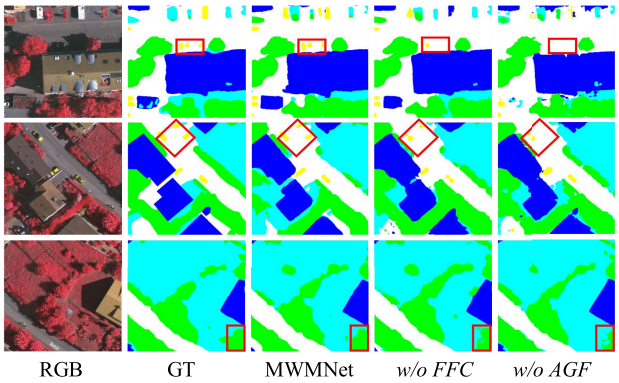


Fig. 6. Comparison of the visual results of the proposed method, *w/o FFC* and *w/o AGF*.

1) Without FFC and AGF Modules (FGMNet): We quantitatively assessed the performance impact of our FFC module through controlled experiments (Table IV).

The ablation studies revealed significant accuracy degradation when the FFC was removed. On the Vaihingen dataset, the mAcc and mIoU decreased by 0.97 and 1.04 percentage points, respectively, whereas on the Potsdam dataset, they decreased by 1.26 and 1.33 percentage points, respectively. By leveraging the phase and amplitude alignment mechanisms, the FFC module captures intrinsic correlations between the RGB and nDSM modalities more effectively than traditional cascade fusion, achieving more accurate classification in regions with blurred boundaries. Fig. 6 shows that removing the FFC resulted in detail loss, demonstrating that the proposed frequency-domain fusion improves cross-modal integration and preserves features.

The ablation studies provided a comprehensive validation of the effectiveness of the AGF module. When this module was removed, we observed consistent performance degradation across both datasets: mAcc decreased by 1.12% and mIoU by 1.18% on Vaihingen, with respective decreases of 1.42% and 1.19% on Potsdam. The underlying reason is that AGF uses a Gaussian prior distribution to normalize feature responses, effectively suppressing noise interference when processing remote sensing images with complex terrain features and shadowing effects. Fig. 6 confirms that AGF leverages this clean feature representation to better focus attention on key areas, thereby improving accuracy.

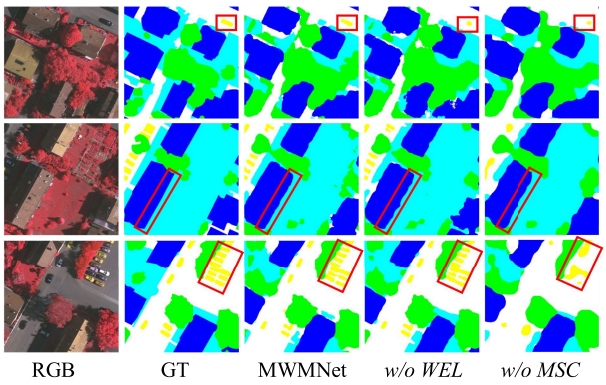


Fig. 7. Comparison of the visual results of the proposed method, *w/o WEL*, and *w/o MSC*.

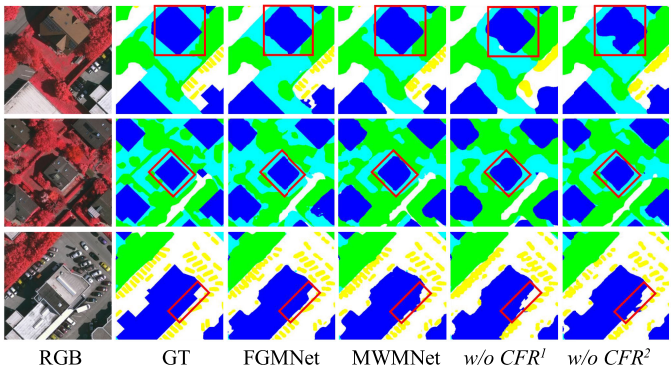


Fig. 8. Comparison of the visual results of our method and *w/o CFR*.

2) Without WEL and MSC Modules (MWMNet): The experimental results revealed consistent performance degradation when removing the WEL module (Table V): the mAcc decreased by 1.09% and the mIoU by 1.13% on Vaihingen, with greater decreases of 1.11% and 0.95%, respectively, on Potsdam. Unlike traditional convolutions, WEL employs a DWT to decompose features into low- and high-frequency subbands, allowing the model to preserve both subtle edges and macrolevel object contours. As shown in Fig. 7, removing WEL causes discontinuous boundaries, indicating that its parallel wavelet design is critical for precise segmentation.

The ablation studies revealed the critical importance of the MSC module. Removing it caused mAcc and mIoU to reduce by 1.05% and 0.98%, respectively, when used on the Vaihingen data and 0.94% and 0.88%, respectively, on the Potsdam dataset. As illustrated in Fig. 7, the absence of the MSC led to attention dispersion and impaired spatial dependency modeling. These results demonstrate the effectiveness of the MSC in enhancing feature discriminability through its position-aware attention mechanism.

3) Without CFR Method: When the feature interaction paths between networks were removed, the performance declined significantly; the mIoU of the two networks decreased by 1.05% and 1.13% on the Vaihingen dataset and by 1.02% and 0.95% on the Potsdam dataset (Table VI), respectively.

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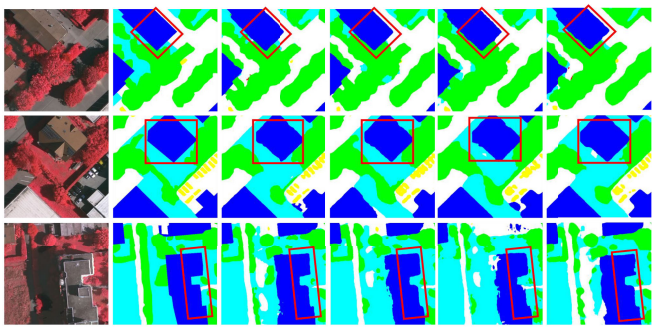


Fig. 9. Comparison of the visual results of our method and *w/o ACC*.

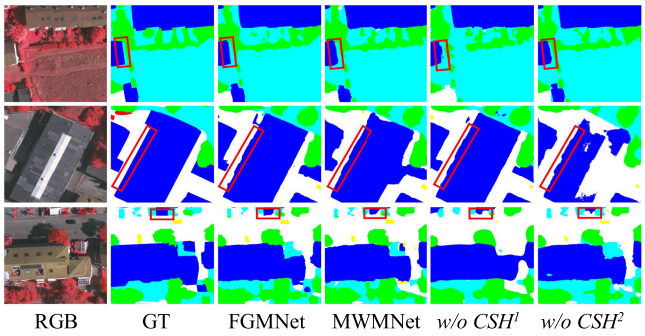


Fig. 10. Comparison of the visual results of our method and *w/o CSH*.

Fig. 8 confirms that the proposed CFR framework successfully aligned the feature distributions between networks, enhancing their complementarity and multiscale extraction capabilities. In these results, *w/o CFR*^{1} denotes the performance of FGMNet with the CFR module removed, whereas *w/o CFR*^{2} denotes the performance of MWMNet with the CFR module removed.

4) Without ACC Mechanism: After disabling the ACC, both models showed a decline in performance (Table VII): on the Vaihingen dataset, the mAcc decreased by 0.88% and 0.95%, respectively, and the mIoU decreased by 0.96% and 0.94%, respectively. On the Potsdam dataset, the mAcc decreased by 1.02% and 1.07%, respectively, and the mIoU decreased by 1.07% and 0.97%, respectively.

As shown in Fig. 9, removing the ACC mechanism weakened the ability of the model to focus effectively on key regions. In these results, *w/o ACC*^{1} denotes the performance of FGMNet with the *ACC* module removed, whereas *w/o ACC*^{2} denotes the performance of MWMNet with the *ACC* module removed.

5) Without CSH Feature Learning: After removing the CSH feature learning in the decoding stage, the difference between the outputs of the two networks increased, whereas the outputs of both models decreased to a certain extent. On the Vaihingen test set, the mAcc of the models decreased by 1.05% and 1.17%, respectively, and the mIoU decreased by 1.13% and 1.20%, respectively. Fig. 10 shows that the absence of this scheme reduced the detail-retention ability and information consistency of the two models. In these results, *w/o CSH*^{1} denotes the performance of FGMNet with the *CSH* module removed, whereas *w/o CSH*^{2} denotes the performance of MWMNet with the *CSH* module removed.

TABLE IX
ABLATION EXPERIMENTS FOR SENSITIVITY ANALYSIS OF LOSS WEIGHTS ON THE VAIHINGEN DATASET

Conditions	CFR	ACC	CSH	FGMNet-CL		MWMNet-CL	
				mAcc	mIoU	mAcc	mIoU
Variant 1	1.5	1.0	1.0	88.84	80.87	88.73	80.47
Variant 2	1.0	1.5	1.0	88.79	80.81	88.66	80.39
Variant 3	1.0	1.0	1.5	88.65	80.69	88.61	80.34
Ours	1.0	1.0	1.0	89.03	81.17	88.91	80.62

6) Without the nDSM Modal Input: The multimodal network provides a global perspective by fusing multisource information and produces smoother feature representations, whereas the unimodal network acts as a dedicated modality expert, delivering specialized predictions that remain unaffected by noise from other modalities. During collaborative training, the unimodal network can implicitly absorb elevation knowledge, and the multimodal network learns the finer grained features, forming a virtuous cycle of mutual distillation and reinforcement.

As shown in Table VIII, removing the nDSM modality reduces the FGMNet to a unimodal network that relies solely on RGB input while retaining the Fourier extraction layer, thereby impairing the overall collaborative training process. The cross-modal complementary chain is thus severed, and the unimodal expert loses its source for implicitly absorbing elevation knowledge, thereby lowering the upper bound for distinguishing similar ground objects. Moreover, the collaboration paradigm shifts from heterogeneous complementarity to homogeneous inputs, easily leading to feature homogenization and redundancy, which weakens the synergistic advantage of the consistency constraint.

7) Sensitivity Analysis on Loss Weights: The monitoring of gradient variation curves and loss function trajectories in the early stages of experimentation revealed that the individual loss terms naturally converged to similar orders of magnitude in the mid-to-late training phase. Therefore, setting all coefficients to 1 ensures a stable joint convergence state and can, to a certain extent, prevent a single optimization objective from dominating the feature learning process of the heterogeneous branches. We evaluated different loss weight combinations on the Vaihingen dataset, and the results are presented in Table IX. When enhancing the influence of a single collaborative component, the final performance exhibits a slight downward change.

E. Cross-Domain Validation on RGB-D Mirror Dataset

To comprehensively evaluate the generalization capabilities of the HetCoLearn framework and its two constituent networks across different application scenarios, we conducted cross-domain experiments on the RGB-D Mirror dataset. We adapted HetCoLearn to the binary segmentation task and employed mean absolute error (MAE) and binary error rate (BER) as the primary evaluation metrics, with IoU used as a supplementary measure for comprehensive assessment.

As shown in Table X, both FGMNet and MWMNet benefit from collaborative learning in this task, consistently

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TABLE X
QUANTITATIVE EVALUATION OF MODELS BASED ON AN RGB-D MIRROR DATASET

Models	IoU↓	BER↓	MAE↓
AsymFromer [59]	62.64	13.48	0.077
MirrorNet [60]	68.37	8.66	0.062
PDNet [61]	77.77	7.77	0.042
SANet [62]	78.43	8.16	0.041
SATNet [63]	78.42	9.31	0.031
FGMNet-CL	78.48	8.24	0.039
MWMNet-CL	78.39	8.51	0.044

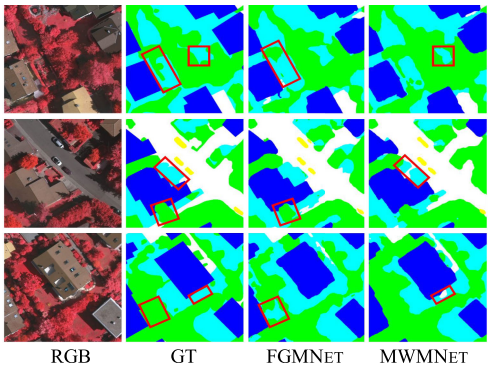


Fig. 11. Comparison of the failure results of ours.

achieving superior performance compared to their independently trained counterparts. Notably, FGMNet achieved the best results among existing mainstream methods, demonstrating that the proposed HetCoLearn framework exhibits strong transferability to computer vision tasks beyond remote sensing. Notably, both FGMNet and MWMNet achieved state-of-the-art performances, demonstrating that the proposed HetCoLearn framework can be generalized effectively to computer vision tasks beyond remote sensing.

F. Failure Cases and Analyses

An in-depth analysis of the proposed HetCoLearn collaborative learning framework revealed that while it significantly enhances the overall performance of individual models, certain mechanisms and categories still exhibit collaborative failures or uneven improvements.

We visualized several experimental failure cases, as shown in Fig. 11. The results show that MWMNet-CL notably underperformed relative to FGMNet-CL in the "Low veg." category. This suggests that, during collaborative learning, the MWMNet may not effectively absorb the features required to encode subtle distinctions between low vegetation and ground surfaces from the FGMNet, thereby limiting the benefits of collaboration. To address this, we designed the ACC mechanism to align the attention responses between the two models. However, the results show that FGMNet-CL continued to underperform in the "Tree" category, indicating that the collaborative process may dilute discriminative attention for certain classes. These cases illustrate that cross-model collaboration does not universally guarantee improvements; its success relies heavily on the structural compatibility and task alignment between models.

In the HetCoLearn framework, while collaborative learning can enhance overall performance, there remains a risk of degradation in specific categories and complex scenarios. Heterogeneous networks tend to exhibit inconsistent optimization directions in the feature space, and bidirectional knowledge transfer can generate conflicting guidance signals at complex boundaries, thereby weakening the edge sensitivity of individual models. In particular, when a certain class of samples suffers from severe noise in a specific modality, consistency constraints may cause erroneous knowledge to propagate to branches that originally performed better, diluting their discriminative attention. As shown in Table I, for the "Tree" category, the FGMNet segmentation accuracy improved marginally from 86.14% to 87.07%. When the FGMNet processes global frequency features via the Fourier transform, if the nDSM data in forested areas contain noise, the resulting consistency constraints may mislead the wavelet-based edge extraction branch of the MWMNet, leading to feature dilution.

To address this issue in the course of future work, we propose to introduce a dynamic weighting mechanism that uses gating functions to control the direction and intensity of knowledge transfer based on local feature reliability, thereby adaptively modulating the collaborative strength. Specifically, when a branch encounters low-quality input or severe modality-specific noise, its prediction confidence drops below a preset reliability threshold, automatically reducing its collaborative loss weight. By transforming the collaboration into a conditional and self-gated interaction mechanism, this approach effectively prevents the propagation of erroneous or noisy information from low-quality samples, thus safeguarding challenging categories such as "low vegetation" and "trees" from causing performance degradation and the dilution of discriminative features.

V. CONCLUSION

In this study, we proposed HetCoLearn, a heterogenous collaborative learning framework that integrates the FGMNet for multimodal RGB–nDSM fusion, the MWMNet for unimodal RGB feature learning, and a three-level collaboration strategy comprising CFR, ACC, and CSH for remote sensing semantic segmentation. Through cross-network collaborative learning, the framework overcomes the limitations of both cross-modal fusion and unimodal feature extraction. Quantitative evaluation results demonstrate that the proposed method achieves significant performance gains on both the Vaihingen and Potsdam datasets. However, the ablation experiments and in-depth analysis of failure cases reveal potential failure modes that emerge during the collaborative training of heterogeneous networks, particularly under severe noise and class-specific ambiguity. Specifically, when the input data contain severe noise, unconstrained consistency guidance at the decoding stage may propagate erroneous knowledge, thereby diluting the discriminative attention of the peer branch and leading to performance degradation for specific classes. This suggests that such HetCoLearn strategies are closely tied to structural compatibility and dynamic task alignment under varying data quality. In light of these findings, future research will focus on developing a dynamic confidence gating mechanism

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that selectively filters knowledge transfer based on real-time prediction confidence, thus preventing the spread of erroneous information from low-quality samples. Furthermore, we plan to explore lightweight asynchronous cross-network information sharing strategies to reduce the computational overhead of synchronous joint training and better-balanced segmentation accuracy with deployment efficiency in practical remote sensing applications.

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